Assignment: RSA Key Validation

Course: Cryptography
Task: Identify which suggested key pairs form valid RSA keys

Introduction

We are given several suggested RSA key tuples of the form:

$$(x,\ y,\ p,\ q,\ n)$$

where:

- x is the public exponent e,
- y is the private exponent d,
- p and q are supposed prime factors of n,
- $n = pq$ is the RSA modulus.

Our job is to check which rows form **valid RSA key pairs**. Each row must satisfy the fundamental RSA requirements:

1. p and q must be **prime**.
2. $n = pq$ must be correct.
3. $\varphi(n) = (p-1)(q-1)$.
4. e must be coprime with $\varphi(n)$:
$$\gcd(e, \varphi(n)) = 1.$$
5. d must be the modular inverse of e modulo $\varphi(n)$:
$$e \cdot d \equiv 1 \pmod{\varphi(n)}.$$

If any condition fails, the row is invalid.

How to Test Whether a Number is Prime

To determine whether a number m is prime, it is **not** necessary to test all numbers up to m. It is sufficient to:

1. Compute $\sqrt{m}$,
2. Test whether any prime number $\leq \sqrt{m}$ divides m.

If no such divisor exists, then m is prime.
Example:

$$2721$$

Since $\sqrt{2721} \approx 52$, we only need to test divisibility by the primes:

$$2,\ 3,\ 5,\ 7,\ 11,\ 13, \ldots,\ 47.$$

Checking:

$$2721 \div 3 = 907$$

This divides evenly (no remainder), meaning:

$$2721 = 3 \times 907$$

Therefore, 2721 is **not prime**.
This method allows fast manual primality testing for all values in the given problem.

Row-by-Row Analysis

Row 1

$$p = 5783,\ q = 6553,\ n = 37895999$$

Primality: Both p and q have no prime factors below their square roots, so both are prime.

$$n = 5783 \times 6553 = 37895999$$

$$\varphi(n) = (5783 - 1)(6553 - 1) = 37883664$$

$$\gcd(36661763,\ 37883664) = 1$$

$$36661763 \times 31000091 \equiv 1 \pmod{37883664}$$

Conclusion: Valid RSA key pair.

Row 2

$$p = 3879$$

Check primality:

$$3879 \div 3 = 1293$$

Since it divides evenly, 3879 is not prime. RSA requires both p and q to be prime.
Conclusion: Invalid (factor is not prime).

Row 3

$$p = 6163,\ q = 4217,\ n = 25989371$$

Both p and q pass primality testing.

$$n = 6163 \times 4217 = 25989371$$

$$\varphi(n) = 25978992, \qquad \gcd(682183,\ 25978992) = 1$$

$$682183 \times 22904887 \equiv 1 \pmod{25978992}$$

Conclusion: Valid RSA key pair.

Row 4

We are given:

$$p = 5477, \quad q = 2437, \quad e = 2\,124\,966$$

1. Primality

Both p and q are confirmed prime.

2. Compute n

$$n = 5477 \times 2437 = 13\,347\,449$$

3. Compute $\varphi(n)$

$$\varphi(n) = (5477 - 1)(2437 - 1) = 5476 \times 2436 = 13\,339\,536$$

4. Check $\gcd(e, \varphi(n))$

We must have:

$$\gcd(e, \varphi(n)) = 1$$

Compute:

$$\gcd(2124966,\ 13339536)$$

5. Euclidean Algorithm

$$\begin{aligned}
13339536 &= 2124966 \times 6 + 589740 \\
2124966 &= 589740 \times 3 + 355746 \\
589740 &= 355746 \times 1 + 233994 \\
355746 &= 233994 \times 1 + 121752 \\
233994 &= 121752 \times 1 + 112242 \\
121752 &= 112242 \times 1 + 9510 \\
112242 &= 9510 \times 11 + 7632 \\
9510 &= 7632 \times 1 + 1878 \\
7632 &= 1878 \times 4 + 120 \\
1878 &= 120 \times 15 + 78 \\
120 &= 78 \times 1 + 42 \\
78 &= 42 \times 1 + 36 \\
42 &= 36 \times 1 + 6 \\
36 &= 6 \times 6 + 0
\end{aligned}$$

Thus:

$$\gcd(2124966, 13339536) = 6$$

6. Conclusion

Since:

$$\gcd(e, \varphi(n)) = 6 \neq 1$$

e and $\varphi(n)$ are not coprime, meaning e has no modular inverse and therefore no valid d can exist.

Conclusion: Invalid RSA key pair.

Row 5

$$p = 6833, \quad q = 6823, \quad e = 36\,784\,094$$

1. Primality

Both p and q are prime.

2. Compute $\varphi(n)$

$$\varphi(n) = (6833 - 1)(6823 - 1) = 6832 \times 6822 = 46\,607\,904$$

3. Check gcd

$$\gcd(36\,784\,094,\ 46\,607\,904) = 2$$

4. Conclusion

Since:

$$2 \neq 1$$

e is not invertible modulo $\varphi(n)$, meaning a valid private key d cannot exist.
Conclusion: Invalid RSA key pair.

Row 6

Check primality:

$$2721 \div 3 = 907$$

Since 3 divides 2721, p is composite.
Conclusion: Invalid, since RSA requires prime factors.

Final Result

Only the following rows produce valid RSA key pairs:

- Row 1
- Row 3

All other rows fail either because:

- p or q is not prime, or
- $\gcd(e, \varphi(n)) \neq 1$, making d invalid.